When LLMs meet External Knowledge

Oct 15th | Tingqi Wang

Integrating the external knowledge into LLMs



User

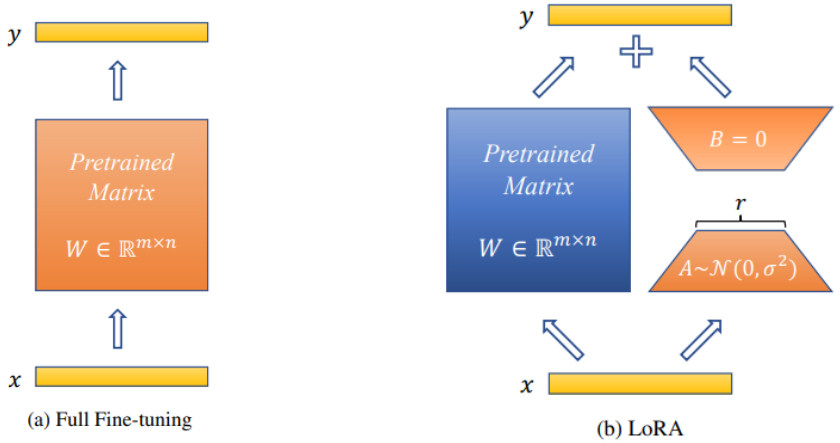
LLM

External Knowledge

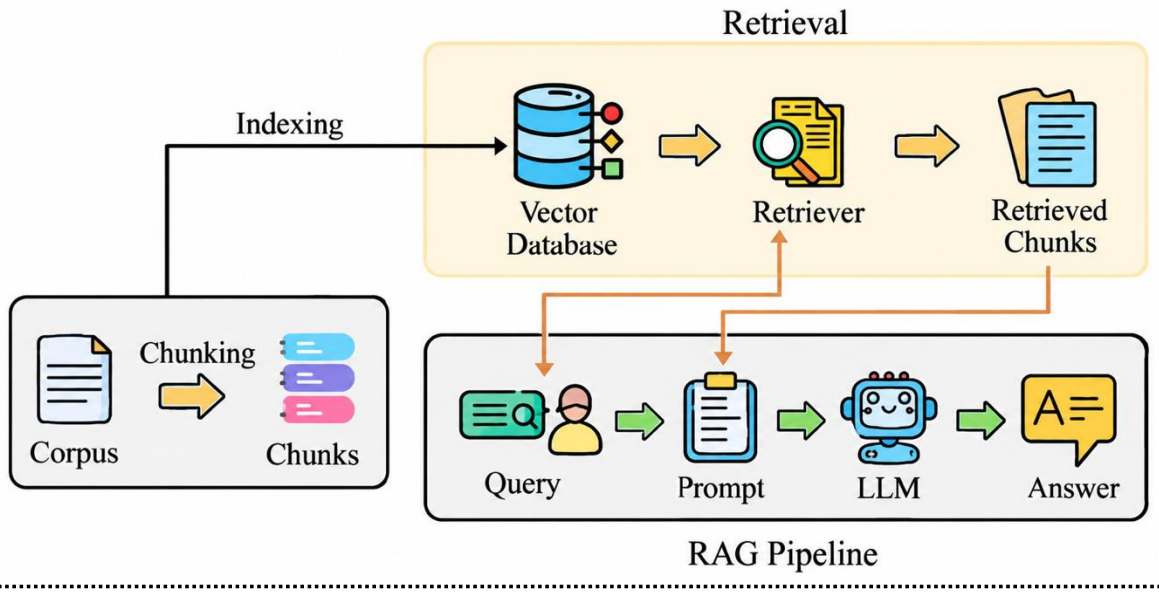
Next Token



Parameter Ajustment



Retrieval-Augmented Generation (RAG)



Agenda

1. FedTAR: Federated Fine-Tuning

Setting: private data distributed across heterogeneous clients.

Challenge: different tasks specialize different experts and can produce incompatible updates.

Detailed discussion

Problem, routing features, subspace alignment, aggregation, and experimental evidence.

2. FuseRAG: Graph-based RAG

Setting: multi-hop question answering over graph evidence.

Challenge: linear contexts must express graph relations while also supporting cache reuse.

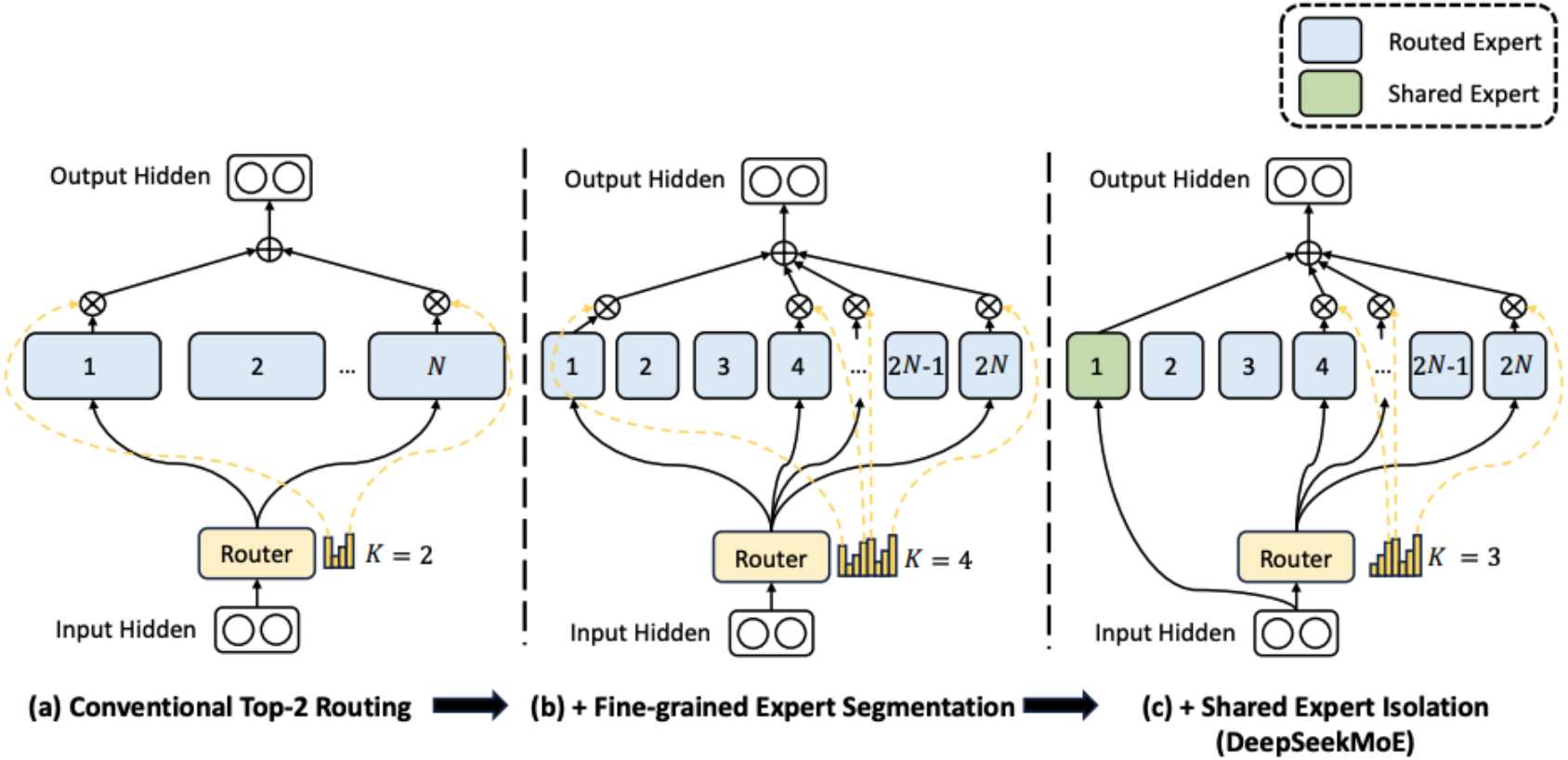
Background-focused discussion

Graph RAG, token order, KV cache, and the motivation for a different evidence interface.

FedTAR: Task-Aware Federated Fine-Tuning for MoE-based Large Language Models

Background: Mixture-of-Experts (MoE) Language Models

Powerful and Cost-Efiicient

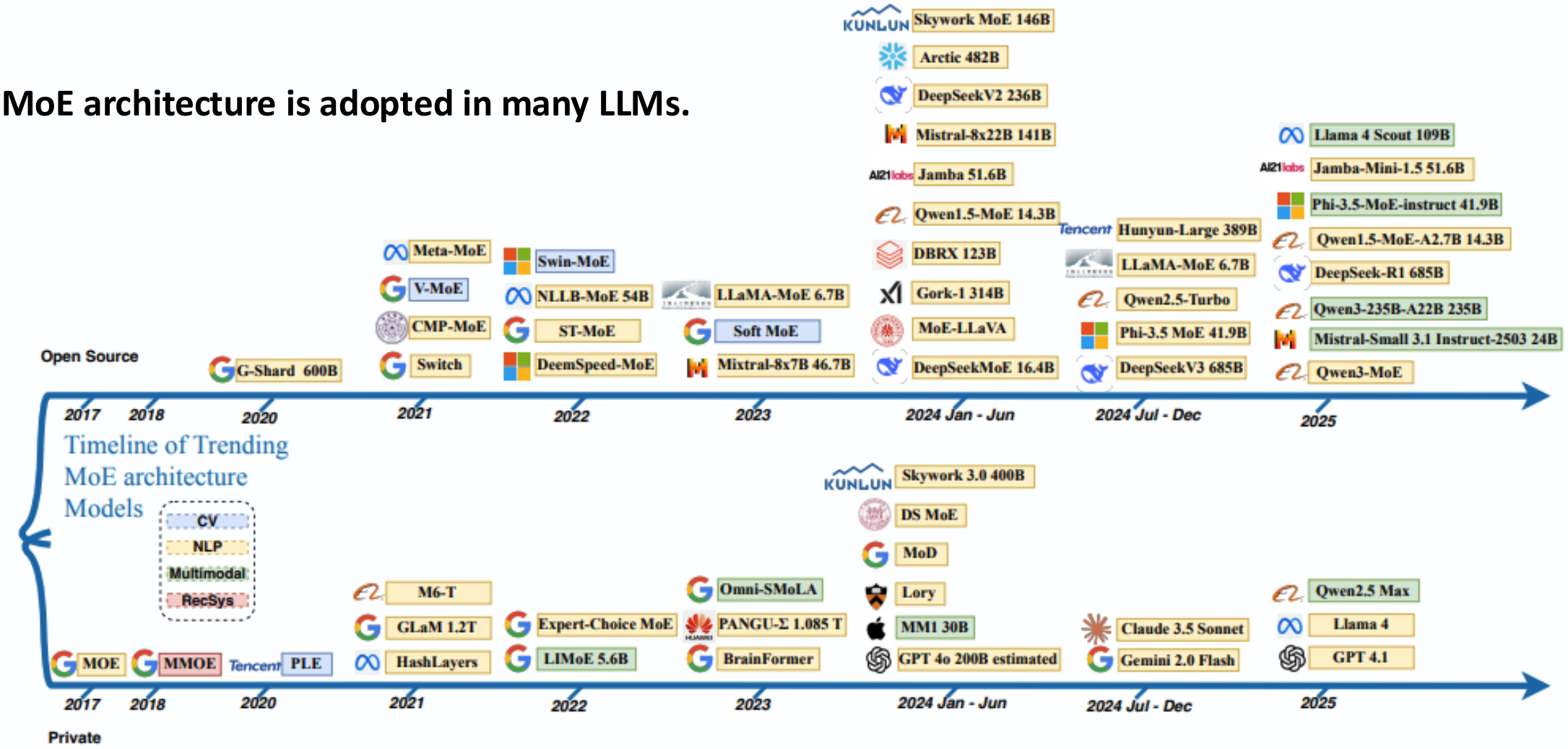


Sparse activation: Each token activates only a small subset of experts, allowing greater model capacity with limited computation.

Input-dependent routing: Different tokens can select different experts according to their hidden representations.

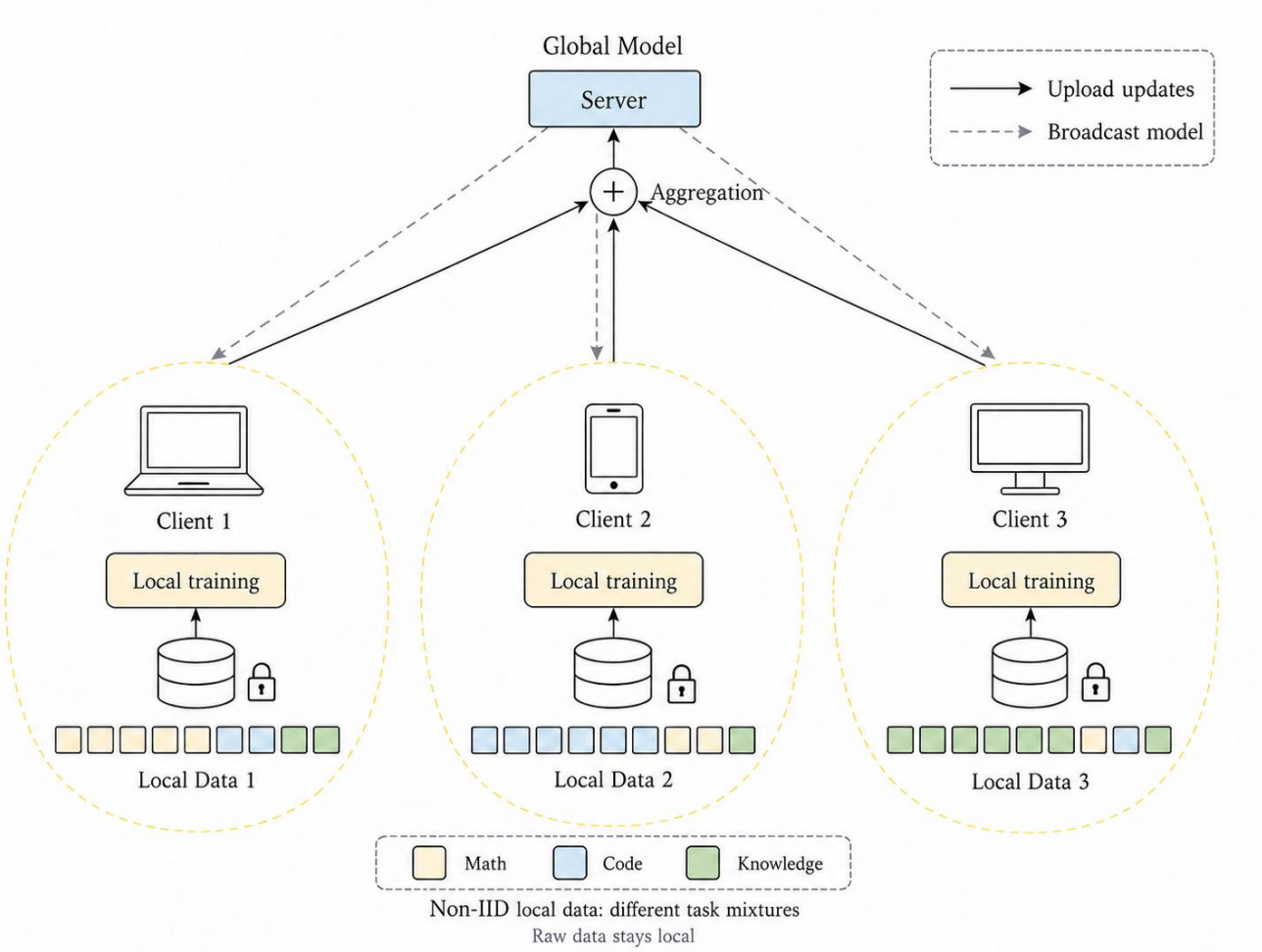
Background: Mixture-of-Experts (MoE) Language Models

MoE architecture is adopted in many LLMs.



Background: Federated Learning with Data Heterogeneity

Devices train locally and exchange model updates to learn a shared global model.



Different local distributions

Devices differ in task preferences, domains, label proportions, and data volume.
Their data are non-IID.

Conflicting local updates

Different local objectives can produce incompatible update directions. Averaging may cancel useful task-specific changes.

Uneven global performance

These conflicts can slow convergence and reduce performance on some client tasks.

Motivation: Challenges in Federated MoE Fine-Tuning

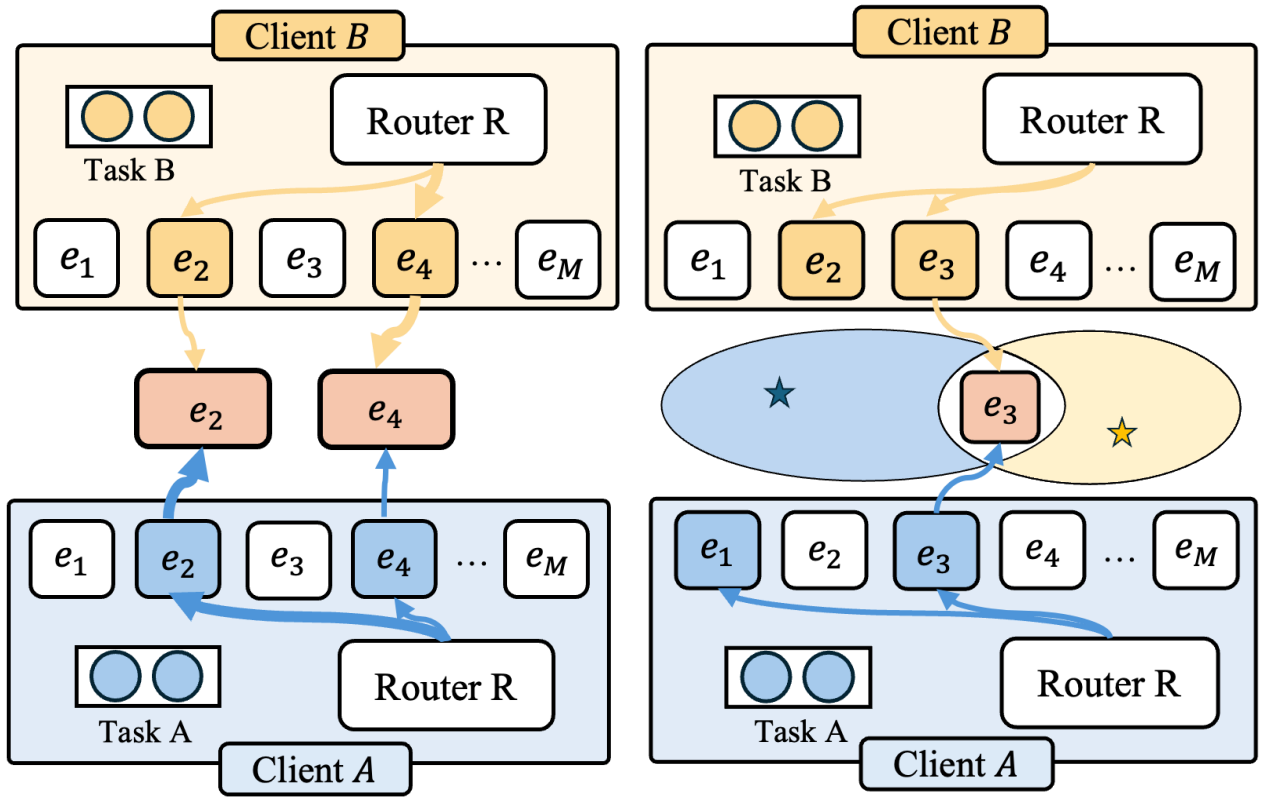
Task differences affect both expert specialization and the directions of shared-expert updates.

1. Weakened specialization

A client may adapt e_2 strongly and e_4 weakly, while another client does the opposite. Uniform averaging reduces these task-specific differences.

2. Shared-expert conflicts

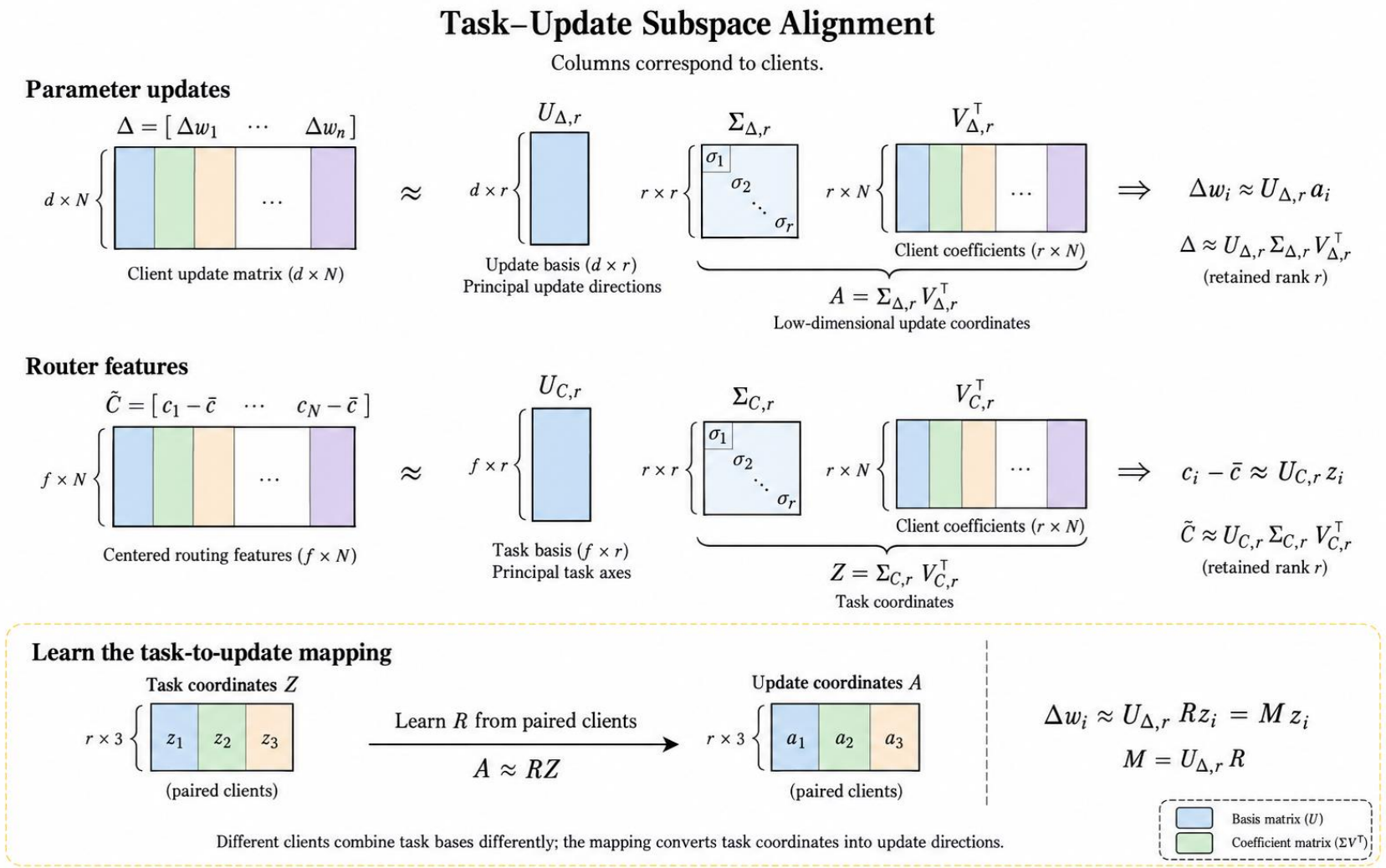
The same expert can serve multiple tasks. Inconsistent optimization directions can cancel useful updates or slow convergence.



How can client updates be aligned with their underlying task intentions in federated MoE fine-tuning?

Insight & Key Idea

Compress the dominant variations before learning a task-to-update relationship.



Update space

Keep the dominant update directions.
Model their task-conditioned variation.

Task space

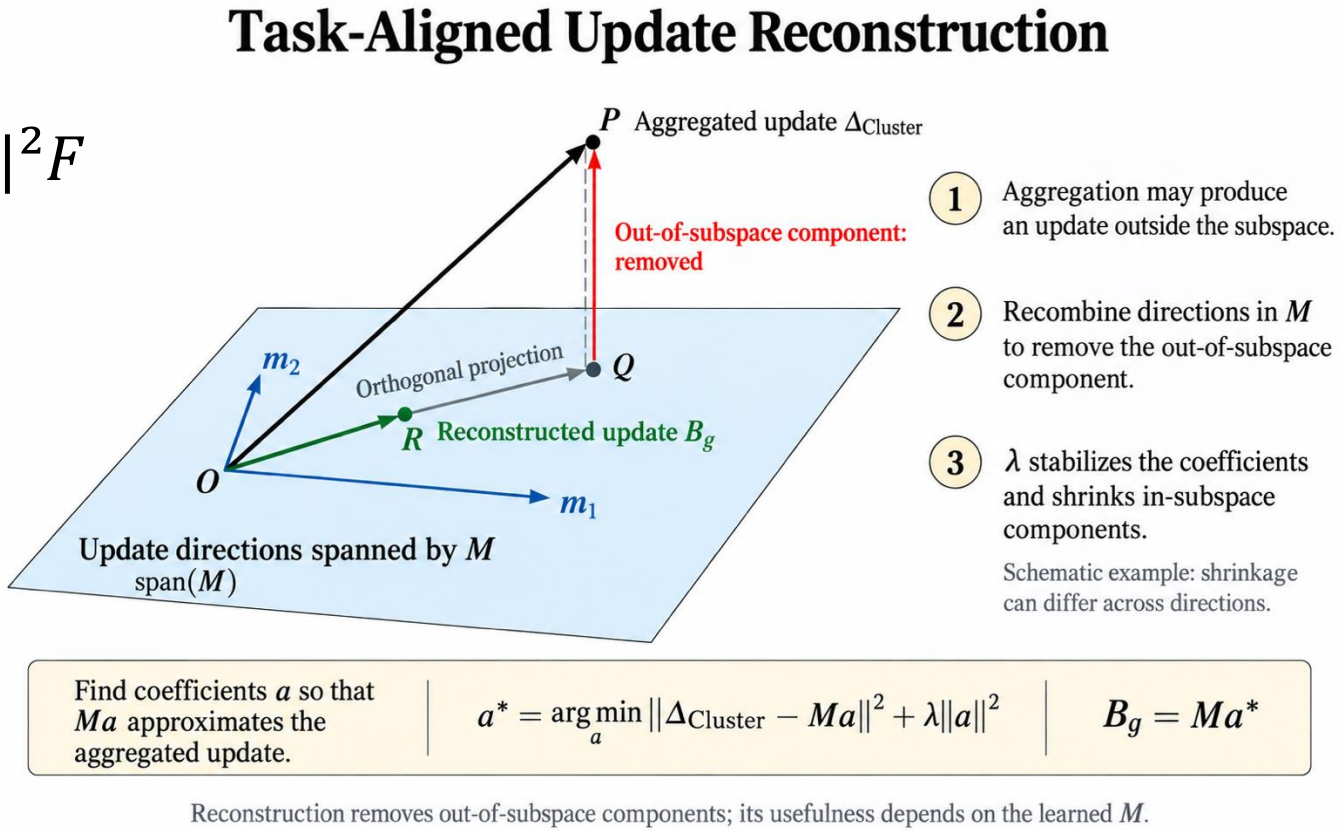
P retains the principal task axes.

Insight & Key Idea

$$B_i \approx Mz_i minM \|\Delta - MZ\|^2 F + \lambda \|M\|^2 F$$

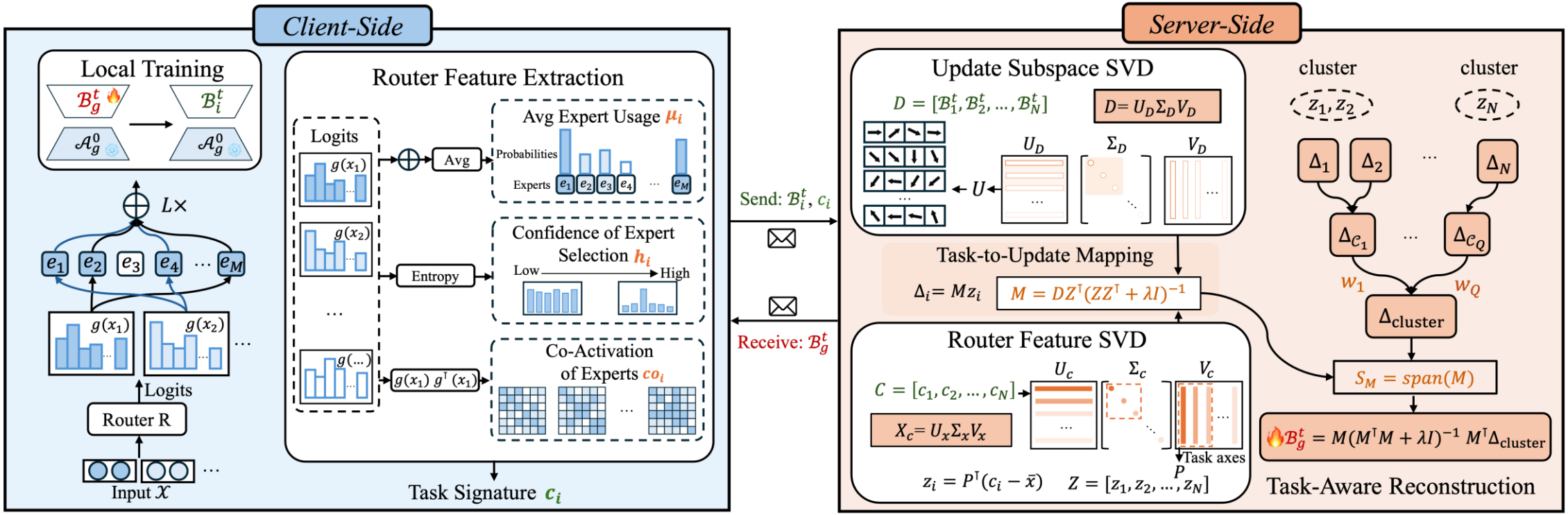
$$M = \Delta Z^T (ZZ^T + \lambda I)^{-1}$$

The column space of M defines the learned task-aligned update subspace.



FedTAR: Method Overview

Routing behavior links local task preferences to the geometry of model updates.



Local LoRA fine-tuning; statistics over router probabilities; upload B_i and c_i.

SVD alignment; within/across-cluster aggregation; task-aligned reconstruction.

Local Updates and Routing Features

Freeze the pretrained model and LoRA A, train B locally, and summarize routing behavior.

Local adaptation

$W = W_0 + BA$
Shared fixed A
Train and aggregate B
Frozen base model

Routing statistic	Definition	What it captures
Mean usage	$\mu_i = E[g(x)]$	Expert preference
Co-activation	$co_i = E[g(x)g(x)^T]$	Expert collaboration
Routing entropy	$h_i = E[-\Sigma g_k \log g_k]$	Selection concentration

Task representation: $c_i = normalize([\mu_i; vec(co_i); h_i])$

The server receives a compact routing signature together with the local update. Raw training examples stay on the client.

Task-Aware Aggregation and Reconstruction

First group compatible tasks, then constrain the combined update to the learned task subspace.

1. Intra-cluster aggregation

Cluster z_i into task group.
Average local updates inside each group.

2. Inter-cluster aggregation

Compute each cluster center.

3. Task-aligned reconstruction

$$Bg = M(M^{\mathrm{T}}M + \lambda I)^{-1}M^{\mathrm{T}}\Delta cluster$$

Bg lies in span(M). Reconstruction removes components outside this subspace and regularizes the retained coefficients.

Communication and Privacy Considerations

Routing features provide extra task information with a limited additional communication cost.

Communication

Standard FL sends an update of dimension d.
The routing signature has $O(K^2)$ dimensions because it includes expert co-activation.
Total per-client cost: $O(d + K^2)$.

Privacy

Routing statistics can still reveal coarse information about local data.
The paper clips the feature vector and adds Gaussian noise before release.
SVD is subsequent post-processing.

Frozen router

Send c_i once. Reuse the task signature across communication rounds.

Trainable router

Refresh c_i each round. Account for repeated releases when assessing privacy.

Experimental Setup

Four tasks, multiple non-IID levels, and strong federated fine-tuning baselines.

Model and federation

Qwen1.5-MoE-A2.7B
14.3B total, 2.7B active parameters
64 experts per layer
10 clients, 100 rounds, 5 local steps
Dirichlet $\alpha \in \{0.1, 0.5, 0.9\}$

Benchmark	Task	Metric
GSM8K	Math reasoning	EM
MMLU	Knowledge	Accuracy
BBH	Complex reasoning	EM
Dolly-15K	Instruction following	ROUGE-L

Training configuration

LoRA rank 8, scaling 16, dropout 0.05
Learning rate 10^{-5}, batch size 16
Maximum sequence length 256 tokens

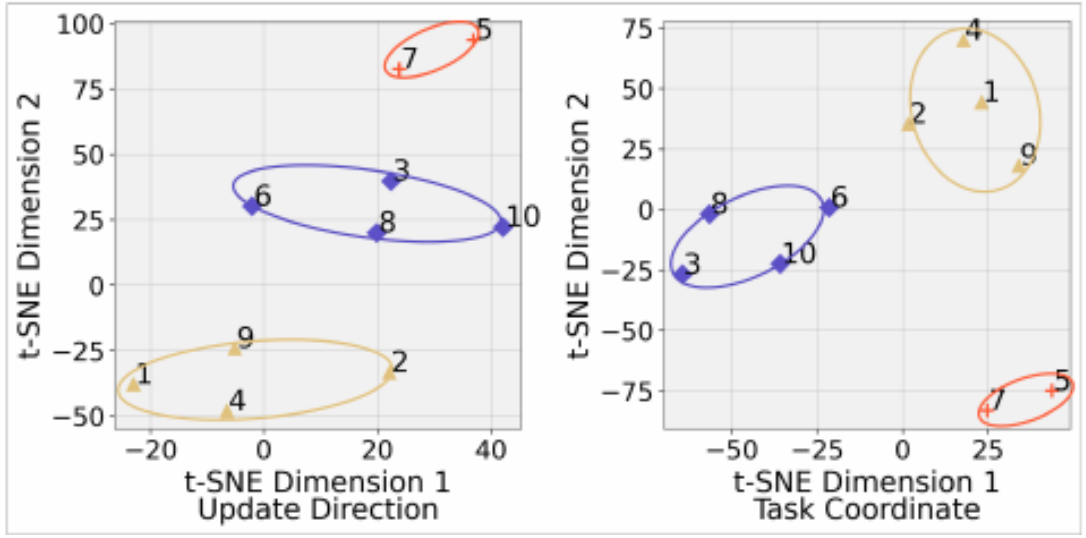
Comparison methods

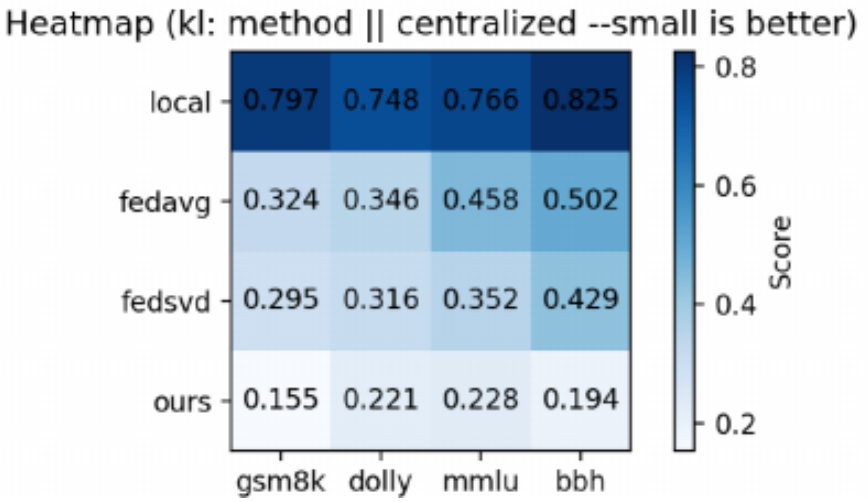
FedAvg, FedProx, SCAFFOLD
FFA-LoRA, FedEx-LoRA, FedSVD

Smaller α means stronger client heterogeneity.

Insight Validation: Task Geometry and Expert Usage

Empirical evidence supports the connection between routing preferences and update behavior.





Task coordinates and updates

Similar client groupings appear in both spaces.
This supports the use of routing statistics to identify task-related update structure.

Expert-usage consistency

FedTAR has the lowest KL divergence on all four benchmarks. Its routing behavior is closer to the centralized reference.

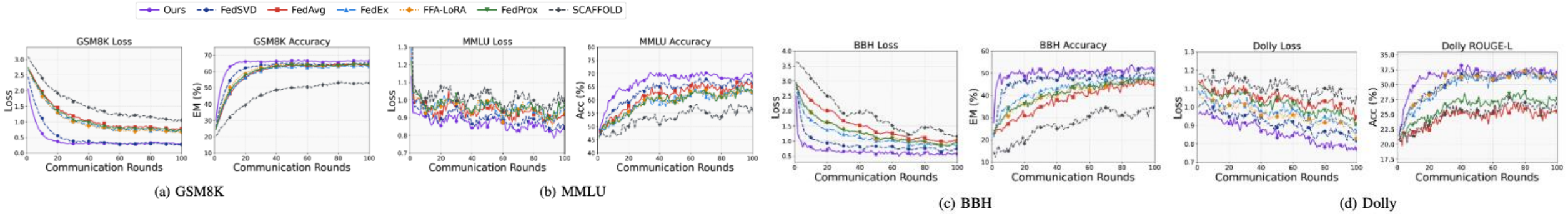
Main Results under Different Heterogeneity

TABLE I: Performance comparison on reasoning and knowledge benchmarks under different task heterogeneity levels. All reported results in the following tables are averaged over three random seeds, with standard deviation below 0.5%.

Method	GSM8K (EM)			MMLU (ACC)			BBH (EM)			AVG.		
α	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
Base Model	61.50%	61.50%	61.50%	62.50%	62.50%	62.50%	36.46%	36.46%	36.46%	53.49%	53.49%	53.49%
Centralized	68.00%	68.00%	68.00%	70.17%	70.17%	70.17%	42.57%	42.57%	42.57%	60.25%	60.25%	60.25%
Local	54.23%	62.01%	64.50%	57.85%	62.10%	62.90%	33.23%	36.60%	37.22%	48.44%	53.57%	54.87%
FedAvg ((AISTATS 2017)) [14]	52.16%	63.87%	66.59%	56.94%	65.00%	64.30%	30.13%	36.72%	39.87%	46.41%	55.20%	56.92%
FedProx (MLSys 2020) [15]	55.38%	63.95%	66.31%	58.26%	63.40%	64.00%	31.76%	37.36%	39.38%	48.47%	54.90%	56.56%
SCAFFOLD (ICML 2020) [16]	48.81%	52.86%	54.28%	54.77%	56.43%	57.32%	28.58%	30.52%	32.51%	44.05%	46.60%	48.04%
FFA-LoRA (ICLR 2024) [22]	62.29%	64.12%	65.03%	61.43%	62.87%	63.41%	36.46%	37.17%	38.08%	53.39%	54.72%	55.51%
Fedex-LoRA (ACL 2025) [23]	63.25%	63.01%	61.00%	61.68%	62.75%	63.57%	33.67%	35.00%	41.35%	52.87%	53.59%	55.31%
FedSVD (NeurIPS 2025) [24]	63.11%	64.62%	66.71%	62.06%	66.22%	65.86%	33.52%	38.21%	39.52%	52.90%	56.35%	57.36%
FedTAR (ours)	64.37%	66.34%	67.43%	64.81%	68.95%	69.20%	37.98%	41.76%	41.95%	55.72%	59.02%	59.53%

TABLE II: Results on dolly (ROUGE-L).

Method	Dolly-15K		
α	0.1	0.5	0.9
Base Model	24.54%	24.54%	24.54%
Centralized	34.17%	34.17%	34.17%
Local	25.89%	27.03%	33.23%
FedAvg	23.56%	25.57%	34.05%
FedProx	24.45%	27.65%	33.17%
SCAFFOLD	23.17%	26.80%	26.40%
FFA-LoRA	30.02%	31.60%	33.61%
Fedex-LoRA	30.58%	31.36%	33.00%
FedSVD	31.21%	31.53%	32.90%
FedTAR (ours)	31.88%	32.12%	34.13%



(a) GSM8K (b) MMLU (c) BBH (d) Dolly

Compared with FedAvg **GSM8K: +12.21 pp** **Dolly-15K: +8.32 pp**

The largest separation appears when client task distributions are highly heterogeneous.

Ablation Study

Every major component contributes to the full model under α = 0.5.

TABLE III: Ablation study of the proposed method across four datasets ($\alpha = 0.5$).

Variant	GSM8K	Dolly	MMLU	BBH	Avg.
Full model	66.34%	32.12%	68.95%	41.76%	52.29%
w/o SVD	64.17%	26.56%	66.24%	37.31%	48.57%
w/o clustering	63.65%	27.01%	67.87%	39.14%	49.42%
w/o reconstruction	64.83%	27.75%	67.43%	39.77%	49.95%
mean-only router	65.26%	31.24%	68.32%	41.34%	51.54%
coact-only router	65.67%	31.58%	68.51%	41.56%	51.83%
w/o router (FedAvg)	63.87%	25.57%	65.00%	36.72%	47.79%

Routing information

Removing the routing representation produces the largest drop.

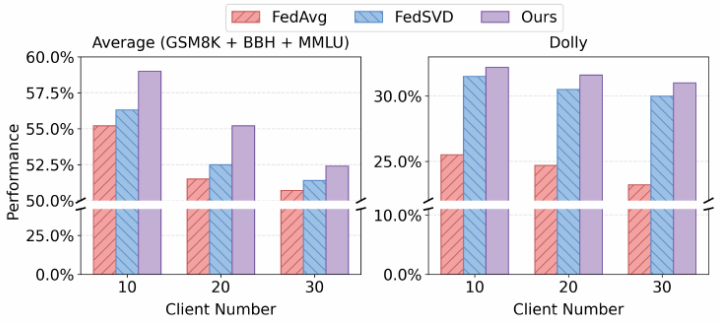
Grouping and projection

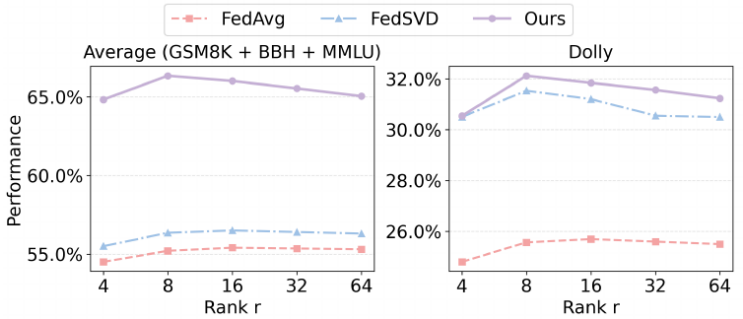
Task grouping helps reduce mixing. Reconstruction adds value beyond grouping alone.

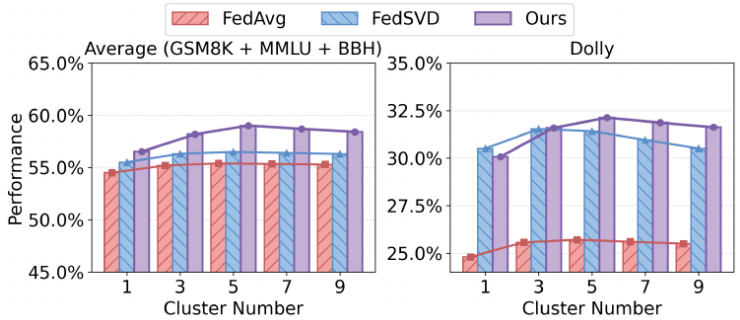
Combining routing statistics is better than using only one statistic.

Robustness to Clients, Rank, and Task Clusters

Sensitivity studies retain the relative advantage while revealing practical trade-offs.







More clients

Overall quality decreases as clients increase.
FedTAR remains ahead of FedAvg and FedSVD in the tested settings.

LoRA rank

Increasing rank from 4 to 8 helps.
Larger ranks bring limited extra gains in the plotted experiments.

Number of clusters

Moderate grouping improves quality.
Too many groups can reduce the amount of data represented by each cluster.

Contributions

Routing structure can guide federated optimization without requiring explicit task labels.

Problem formulation

Identify weakened expert specialization and conflicting shared-expert updates under heterogeneous client tasks.

Method

Represent tasks through routing statistics, align task/update subspaces, aggregate by task groups, and reconstruct global updates.

Experimental evidence

Four benchmarks and several non-IID levels.
Consistent gains over federated baselines.
Ablations support each main component.

Scope of the conclusion

Evidence comes from the evaluated model and datasets. The method relies on informative routing signatures and a useful low-dimensional task–update relationship.

FuseRAG: Fusing Structure and Evidence into Inference for Cost-Efficient Graph-based RAG

Background: RAG and Graph-based RAG

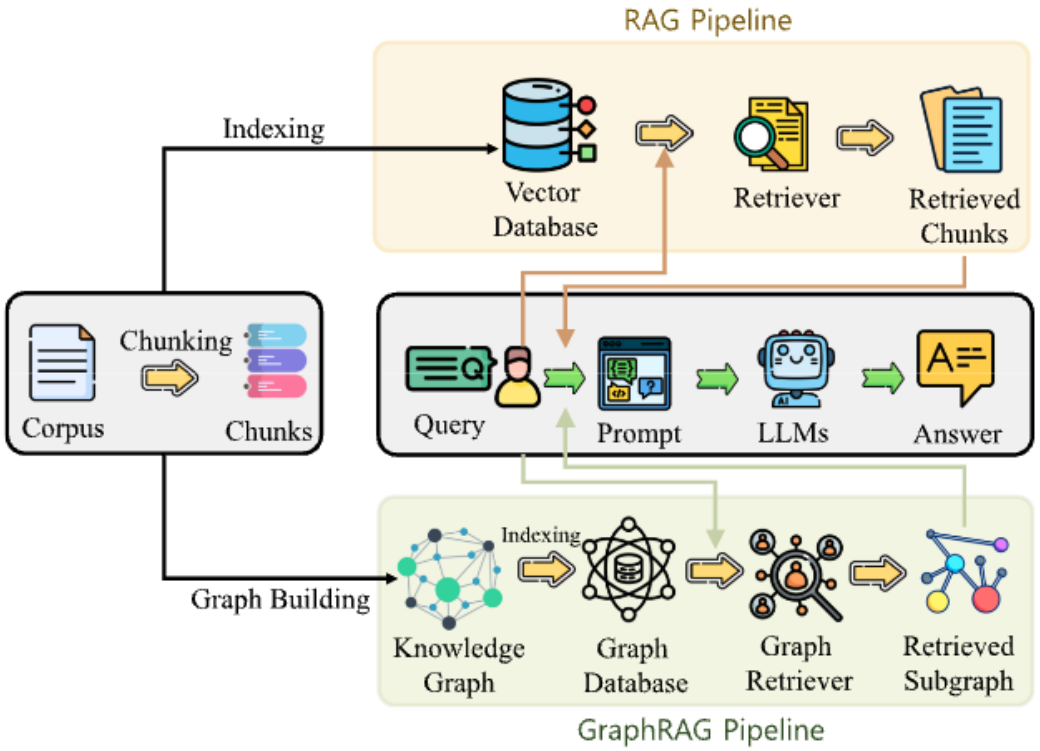
Multi-hop questions require evidence relationships in addition to individually relevant passages.

Why retrieval is needed

Pretraining may omit domain-specific or recently updated knowledge.
RAG supplies external evidence at inference time.

What graphs add

Nodes and edges represent entities and relationships.
A retrieved subgraph can expose paths needed for multi-hop reasoning.



The next challenge is how the retrieved graph enters the LLM.

Background: Prefill, Decoding, and KV Reuse

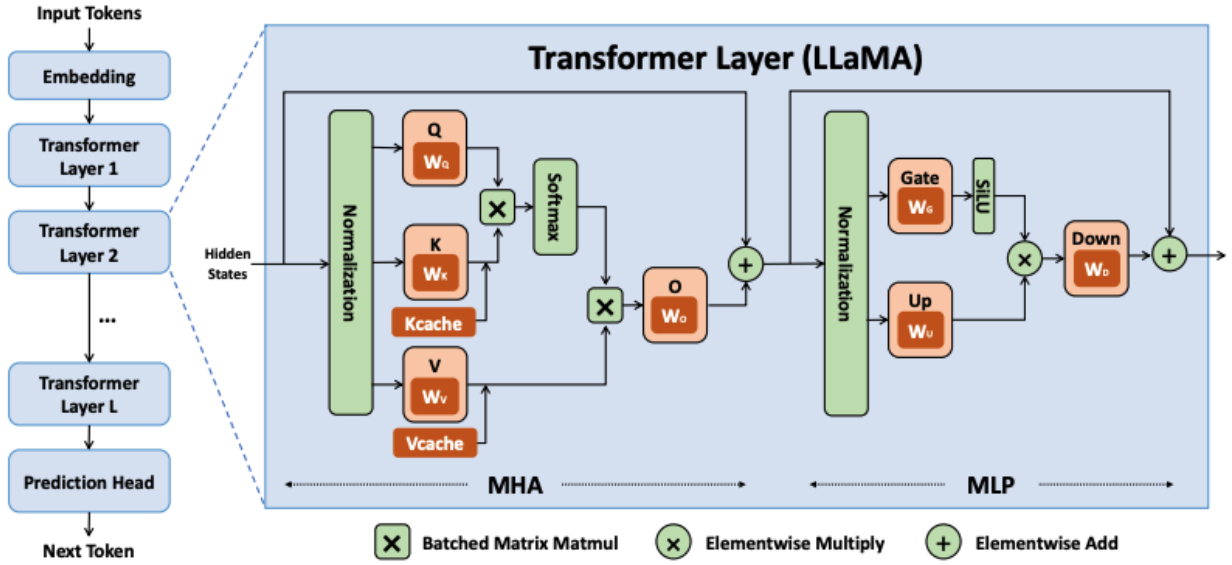
Long RAG contexts make input processing a substantial part of inference cost.

Prefill

Process input tokens in parallel.
Compute and store layer-wise key/value states.
Long retrieved contexts increase this cost.

Decoding

Generate one token at a time using the KV states of earlier tokens.



Across requests

An identical token prefix can reuse earlier KV computation.
Reordering the same evidence can remove this opportunity.

Graph Linearization Changes Evidence Utilization

The same retrieved evidence can produce different reasoning behavior after serialization.

A multi-hop question

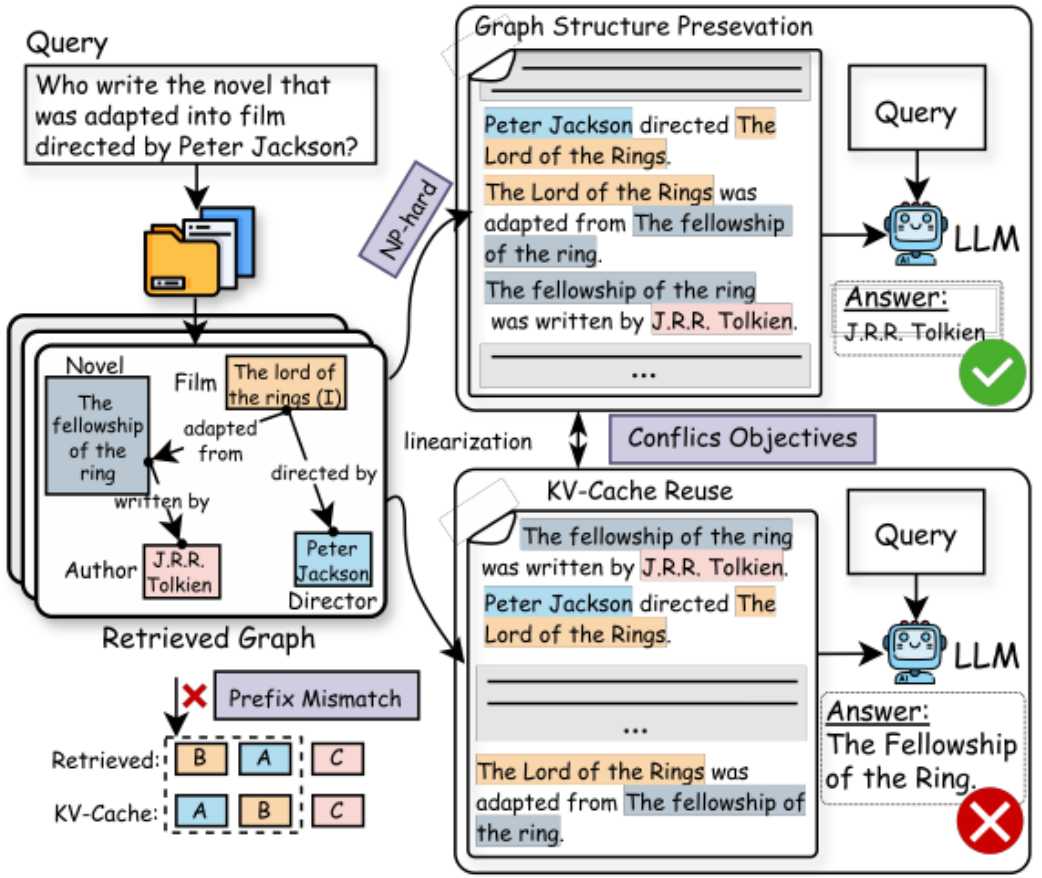
Who wrote the novel adapted into a film directed by Peter Jackson?

The answer depends on following relations across several pieces of evidence.

Two serialization choices

A path-aligned order keeps connected facts adjacent.
A different order can separate the same facts with irrelevant text.

Facts remain available, but their relational locality changes.



The Difficulty of Optimizing Serialization

Even the graph-structure ordering objective alone contains an NP-hard problem.

Structural cost

$$C_{str}(\pi) = \Sigma_{(u,v)} w_{uv} \, |pos_{\pi}(u) - pos_{\pi}(v)|$$

Cache-reuse benefit

$$R_{cache}(\pi) = \max_c LCP(X_{\pi}, X_c)$$

What the optimization asks

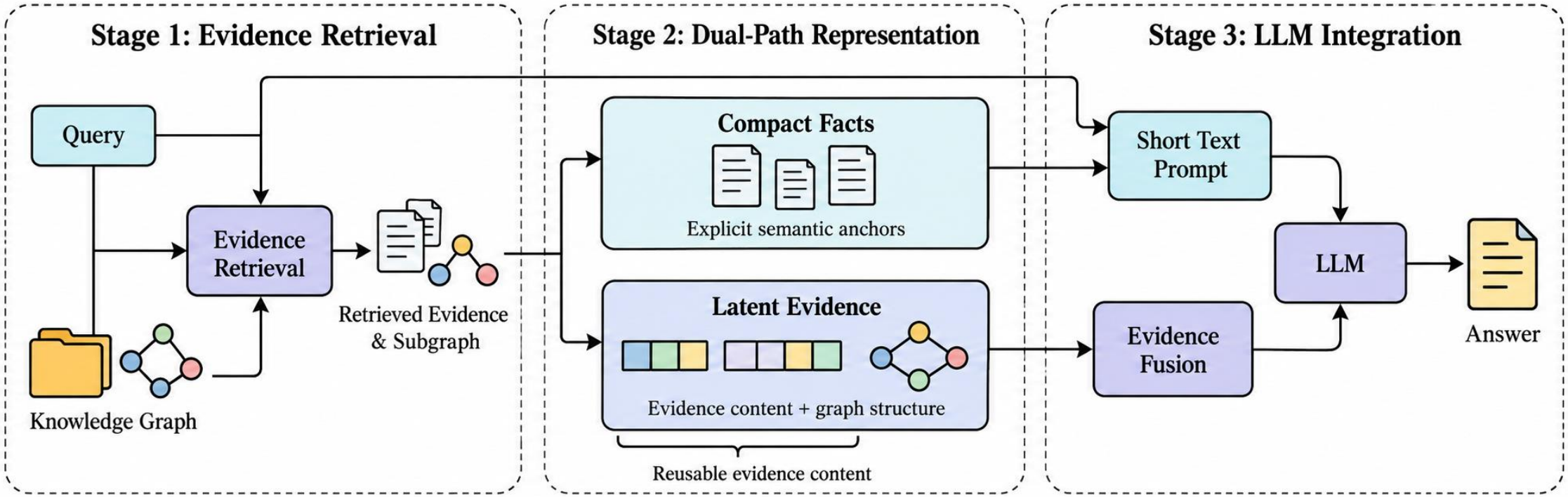
Keep related nodes nearby while matching earlier token prefixes. These goals can favor different orders.

Motivation for FuseRAG

Represent most graph evidence in latent space, so long textual serialization carries less of the burden.

FuseRAG: High-Level Framework

Separate reusable evidence content from query-aware graph interactions; retain a few explicit facts.



Most evidence is represented in latent space; a few explicit facts provide semantic anchors.

1. Retrieve evidence

Keep the relevant subgraph and its evidence units.

2. Represent evidence

Latent evidence and graph interactions, plus compact factual anchors.

3. Integrate with the LLM

Fuse latent memory into attention while anchors enter the text prompt.

FuseRAG: Quality and Latency Overview

TABLE II: QA performance and time-to-first-token (TTFT) across three datasets.

	Method	MuSiQue F1 (%)↑	MuSiQue Acc. (%)↑	MuSiQue TTFT (ms)↓	HotpotQA F1 (%)↑	HotpotQA Acc. (%)↑	HotpotQA TTFT (ms)↓	2Wiki F1 (%)↑	2Wiki Acc. (%)↑	2Wiki TTFT (ms)↓
Llama-3-8B	Zero-shot	7.21 ± 1.32	5.52 ± 0.94	21.07	29.34 ± 2.87	26.53 ± 1.24	22.25	18.88 ± 1.76	33.78 ± 1.60	19.97
	VanillaRAG	12.28 ± 1.14	14.21 ± 1.43	86.23	40.83 ± 2.09	53.87 ± 2.78	89.52	34.22 ± 2.54	41.06 ± 2.87	92.75
	KGP	8.13 ± 0.26	9.43 ± 0.80	3291.84	22.48 ± 0.42	23.57 ± 0.67	3467.37	15.78 ± 1.37	25.84 ± 1.73	3753.40
	ToG	3.86 ± 0.78	5.25 ± 0.53	4053.57	3.76 ± 0.67	28.79 ± 1.76	4267.35	6.37 ± 0.89	42.84 ± 2.48	4553.16
	RAPTOR	10.42 ± 1.21	16.76 ± 1.33	107.25	34.46 ± 1.57	57.12 ± 1.11	110.48	19.62 ± 1.71	45.42 ± 1.64	132.55
	G-Retriever	12.25 ± 1.28	14.76 ± 1.57	2916.84	35.78 ± 2.31	53.56 ± 2.78	3137.67	24.64 ± 2.87	44.58 ± 2.41	3266.33
	LGraphRAG	9.46 ± 2.52	14.77 ± 2.68	167.35	24.81 ± 1.63	43.90 ± 1.78	189.72	15.23 ± 2.02	42.34 ± 2.14	213.15
	FastGraphRAG	18.24 ± 2.53	17.68 ± 2.78	973.79	29.42 ± 2.41	45.76 ± 3.21	1032.65	18.35 ± 2.37	29.89 ± 2.56	1075.30
	LLightRAG	7.91 ± 1.09	6.89 ± 1.48	971.23	25.66 ± 1.62	36.72 ± 1.81	962.46	17.47 ± 1.35	33.85 ± 1.56	968.57
	GLightRAG	5.25 ± 0.77	5.73 ± 0.96	984.69	17.48 ± 1.12	27.72 ± 1.49	979.16	11.67 ± 1.56	23.66 ± 1.73	987.35
	HLightRAG	8.57 ± 0.71	9.24 ± 0.84	1013.77	25.93 ± 1.31	38.61 ± 1.64	1214.25	14.32 ± 1.48	28.60 ± 1.79	1265.32
	GFM-RAG	29.67 ± 1.26	28.64 ± 1.34	2721.32	61.92 ± 1.48	63.26 ± 1.72	2836.15	44.28 ± 1.75	45.80 ± 1.66	2987.57
	KET-RAG	24.25 ± 2.32	22.51 ± 2.88	1186.14	51.75 ± 2.57	53.16 ± 3.14	1212.31	27.90 ± 2.79	41.40 ± 3.43	1394.28
	Clue-RAG-0.5	29.27 ± 1.40	28.32 ± 1.83	4219.46	55.96 ± 1.73	62.08 ± 1.56	4524.77	35.83 ± 1.64	48.20 ± 1.79	4667.38
	Clue-RAG-1.0	28.56 ± 2.30	28.70 ± 2.81	3463.22	56.74 ± 2.66	61.95 ± 2.76	3626.41	36.53 ± 2.52	50.40 ± 3.20	3789.67
	Ours-32	23.23 ± 1.04	24.17 ± 1.62	**58.13**	51.19 ± 1.92	59.38 ± 2.71	**44.26**	38.95 ± 2.94	48.67 ± 3.13	**47.37**
	Ours-64	30.40 ± 1.58	28.72 ± 2.03	61.48	58.92 ± 2.84	62.64 ± 3.10	48.91	45.73 ± 3.03	41.50 ± 3.10	51.87
	Ours-128	**41.08** ± 1.74	**34.50** ± 2.52	65.74	**63.05** ± 2.74	**64.12** ± 3.26	57.13	**55.35** ± 2.98	**51.70** ± 3.20	62.25
Qwen3.5-9B	Zero-shot	7.90 ± 0.53	5.94 ± 0.77	77.23	19.77 ± 1.02	20.20 ± 1.34	75.37	23.64 ± 1.64	24.62 ± 1.60	62.26
	VanillaRAG	30.71 ± 1.24	25.40 ± 1.45	166.54	59.66 ± 2.03	51.30 ± 2.31	176.22	46.12 ± 1.79	48.60 ± 1.64	181.67
	KGP	23.62 ± 1.58	16.91 ± 1.12	4991.32	41.74 ± 3.06	30.55 ± 2.85	5433.74	31.61 ± 2.88	30.04 ± 2.84	5666.83
	ToG	8.98 ± 0.36	12.10 ± 1.53	5029.18	8.88 ± 0.81	35.64 ± 2.97	5294.68	11.49 ± 1.98	49.69 ± 3.10	5655.44
	RAPTOR	22.76 ± 2.62	23.13 ± 3.53	247.36	45.57 ± 3.09	60.92 ± 3.02	255.53	28.24 ± 2.79	47.82 ± 3.19	306.28
	G-Retriever	22.87 ± 1.74	22.25 ± 1.63	3331.69	55.04 ± 3.25	60.54 ± 2.78	4916.75	40.47 ± 3.04	48.78 ± 3.07	4930.02
	LGraphRAG	9.12 ± 0.67	32.60 ± 2.85	278.92	44.07 ± 3.12	50.88 ± 2.87	297.55	31.06 ± 2.68	46.54 ± 3.02	322.89
	FastGraphRAG	28.86 ± 1.53	25.16 ± 1.78	1204.88	48.68 ± 3.62	52.74 ± 3.01	1618.57	34.18 ± 2.94	41.49 ± 3.05	1623.67
	LLightRAG	23.04 ± 1.09	19.30 ± 1.48	1505.72	55.75 ± 2.87	50.20 ± 3.05	1513.67	46.47 ± 1.31	45.50 ± 1.54	1465.92
	GLightRAG	18.53 ± 1.47	16.10 ± 2.17	1513.67	36.74 ± 2.89	34.70 ± 2.95	1525.41	31.61 ± 1.35	40.10 ± 1.48	1485.98
	HLightRAG	28.86 ± 1.22	25.85 ± 1.64	1636.49	46.22 ± 3.14	55.34 ± 3.06	1960.91	34.61 ± 2.99	45.21 ± 3.03	2043.28
	GFM-RAG	27.96 ± 1.45	33.20 ± 2.03	2676.18	60.21 ± 3.03	67.82 ± 2.90	1692.36	42.57 ± 3.18	50.36 ± 3.26	1782.18
	KET-RAG	18.29 ± 2.57	22.06 ± 3.21	1398.55	36.53 ± 2.98	31.66 ± 2.87	1838.76	38.42 ± 3.01	45.88 ± 3.09	2114.42
	Clue-RAG-0.5	39.87 ± 3.56	35.80 ± 4.27	4926.88	62.46 ± 2.91	53.71 ± 2.83	5075.57	45.26 ± 3.00	50.16 ± 3.08	5131.92
	Clue-RAG-1.0	40.62 ± 3.21	36.20 ± 4.10	5027.59	60.75 ± 2.59	55.40 ± 3.10	4326.24	50.36 ± 2.89	46.60 ± 3.10	4487.03
	Ours-32	37.59 ± 1.78	29.45 ± 1.84	**164.57**	41.35 ± 2.78	33.60 ± 3.00	**141.82**	33.56 ± 2.78	29.70 ± 2.80	**143.13**
	Ours-64	41.94 ± 2.27	34.60 ± 2.52	168.45	53.22 ± 2.82	45.60 ± 3.20	143.78	39.71 ± 2.87	36.30 ± 3.00	149.86
	Ours-128	**44.64** ± 2.76	**37.90** ± 2.91	174.52	**64.51** ± 2.72	**56.20** ± 3.20	147.49	**51.63** ± 2.93	**48.80** ± 3.10	154.77

Llama-3-8B on HotpotQA: +28.59 pp F1 and approximately 1.93× TTFT speedup.

Study scope: Llama-3-8B and Qwen3.5-9B; MuSiQue, HotpotQA, and 2Wiki; explicit anchor budgets of 32 / 64 / 128 tokens.

FuseRAG: Quality and Latency Overview

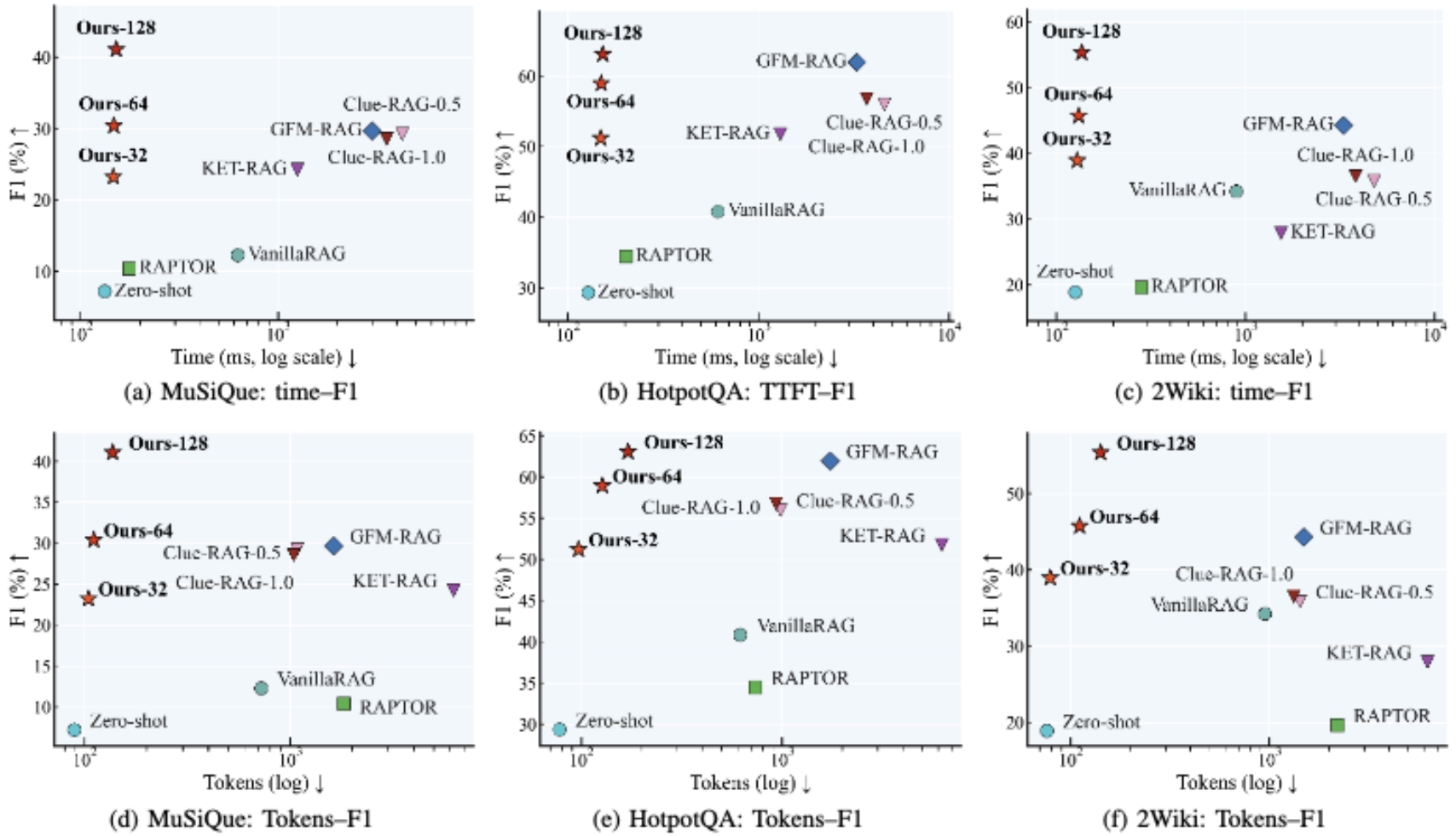


Fig. 3: **QA quality versus inference cost using Llama-3-8B.** The horizontal axes show TTFT or token cost on a logarithmic scale, and the vertical axes show F1. The upper-left region indicates higher quality with lower cost. Stars denote our methods.

Conclusion: Structure in Adaptation and Inference

The two papers address different stages of the LLM lifecycle.

Aspect	FedTAR	FuseRAG
Stage	Federated fine-tuning	Graph-based RAG inference
Main challenge	Heterogeneous expert updates	Graph serialization and KV reuse
Structure used	Routing-derived task coordinates	Relations in retrieved subgraphs
Core idea	Task-aware aggregation and reconstruction	Latent evidence with compact anchors

Shared perspective

Preserve useful structural information when combining client updates or integrating retrieved evidence.

Thanks for Listening!
Q & A